Franklin Minh Nguyen

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Education

Massachusetts Institute of Technology — Cambridge, MA
B.S. in Computer Science and Molecular Biology — *September 2021 – June 2025*

Work Experience

Medtronic — Boston, MA
Software Engineer I — *June 2025 – Present*

- Develop safety-critical GUI features for the **Hugo Robotic-Assisted Surgery System (C++, Qt/QML, Model-View-Controller Architecture)**, supporting clinical workflows used in over **50,000 procedures**
- Collaborate cross-functionally with systems, user-centered design, and requirements engineers to deliver key features including alarm priority indicators, error-recovery workflows, and volume controls
- Own and scale an enterprise string management tool used by **200+ engineers**, automating configuration and QML updates to eliminate legacy bottlenecks across **300+ embedded strings**
- Automated local test environment configuration using Python scripts, cutting setup time by **75%**
- Authored EARS requirements and built automated test suites using **Python/PyTest** to validate software behavior
- Evaluated **Flutter/Dart** against legacy **Qt/QML and C++** via rapid prototyping and a **Pugh decision matrix** to analyze development speed and cost trade-offs for for next-generation system software
- Design **skills for AI agents** that automate redundant development workflows to accelerate feature delivery

Software Engineer Intern — *June 2024 – August 2024*

- Implemented controls to simulate alarms, instrument connections, and ergonomic configurations, utilizing **threads and listeners** to model real-time subsystem communication
- Restructured **4,200+ lines** of legacy Python/PyQt into a modular architecture, improving code maintainability and feature velocity for **25+ engineers**

MIT Urban Risk Lab — Cambridge, MA
Advanced Undergraduate Researcher — *September 2024 – May 2025*

- Developed an inference-based flood propagation model in **Python** using terrain data to predict urban flood spread in low-data environments with **92% validated precision**
- Optimized **GPT-4o Vision** output for classifying urban street conditions through iterative **prompt engineering**, achieving a **34% increase in accuracy** to automate large-scale machine learning dataset annotation

Undergraduate Developer — *February 2024 – June 2024*

- Prototyped a **responsive web platform** using **React.js**, **JavaScript**, **HTML/CSS**, and **the Google Sheets API** to accelerate discovery for climate/energy roles and automate listing ingestion

MIT EECS — Cambridge, MA
Lab Assistant — *August 2023 – April 2025*

- Mentored **150+ computer science students**, providing technical guidance on **Object-Oriented Programming (OOP)**, **recursion**, and **algorithmic thinking**
- Created solution sets and reviewed assignment specifications for **10+ problem sets** across three CS courses

Projects

Shelfie (`myshelfie.onrender.com`) — *January 2024*

- Built a full-stack **MERN** social platform for readers, integrating the **Google Books API** for content discovery
- Won **Most Innovative UI Feature** among 89 teams at MIT's web.lab for dynamic book visualizations

Skills

Languages:	Python, TypeScript, JavaScript, C++, HTML/CSS, SQL, R, QML, Dart, Julia
Frameworks & Libraries:	React.js, Qt, PyQt, Flutter, Express.js, Next.js, RESTful APIs, MUI
Developer & Testing Tools:	Git, GitHub Copilot, Linux, Docker, PyTest, AWS (S3, EC2, IAM), Squish, MongoDB
Productivity & Design:	Atlassian Suite, Figma, Adobe Creative Suite, Polarion, Jama, Miro, LaTeX
Certificates:	Lean Six Sigma Green Belt, C++ Programming Fundamentals (Microsoft), AWS Cloud Technical Skills (AWS)